High dietary fiber intake prevents stroke at a population level.

BACKGROUND & AIMS: This research was aimed at clarifying whether high dietary fiber intake has an impact on incidence and risk of stroke at a population level. METHODS: In 1647 unselected subjects, dietary fiber intake (DFI) was detected in a 12-year population-based study, using other dietary variables, anagraphics, biometrics, blood pressure, heart rate, blood lipids, glucose, insulin, uricaemia, fibrinogenaemia, erytrosedimentation rate, diabetes, insulin resistance, smoking, pulmonary disease and left ventricular hypertrophy as covariables. RESULTS: In adjusted Cox models, high DFI reduced the risk of stroke. In analysis based on quintiles of fiber intake adjusted for confounders, HR for incidence of stroke was lower when the daily intake of soluble fiber was >25 g or that of insoluble fiber was >47 g. In multivariate analyses, using these values as cut-off of DFI, the risk of stroke was lower in those intaking more that the cut-off of soluble (HR 0.31, 0.17-0.55) or insoluble (HR 0.35, 0.19-0.63) fiber. Incidence of stroke was also lower (-50%, $p < 0.003$ and -46%, $p < 0.01$, respectively). CONCLUSIONS: Higher dietary DFI is inversely and independently associated to incidence and risk of stroke in general population.